

19. Gravitation



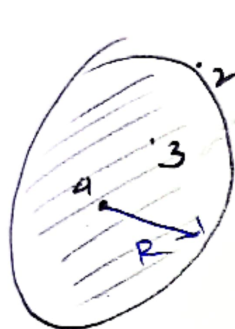
$$g = \frac{GM}{R^2}$$

1. With height: $g' = \frac{g}{(1 + \frac{h}{R})^2}$

2. With Depth:

$$g' = g(1 - \frac{2h}{R})$$

$$g' = g(1 - \frac{d}{R}) \quad g' = \frac{GM}{R^2}$$



$$1. g' = \frac{GM}{R^2}$$

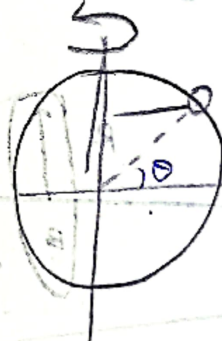
$$2. g' = \frac{GM}{R^2}$$

$$3. g' = \frac{GM}{R^3}(r)$$

$$4. g' = 0$$

3. With Rotation:

$$g' = g - R\omega^2 \cos \theta$$



Kepler's:

i. 1st Law: Elliptical Path.

ii. 2nd Law: Equal Areas $\frac{dA}{dt} = \text{constant}$

iii. 3rd Law:

$$T^2 \propto a^3$$

where

$T \rightarrow$ Time Period of Planet.

$a \rightarrow$ Semi Major Axis.

$$T = \frac{2\pi}{\omega}$$

where $T \rightarrow$ Time Period

$\omega \rightarrow$ Angular Velocity of Planet.

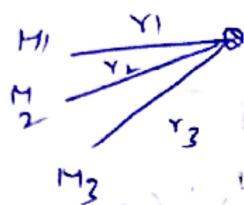
Gravitational Field $\vec{F}$:

1. $\vec{F}$ due to Point Mass:



$$\vec{F} = \frac{GM}{r^2}$$

2.



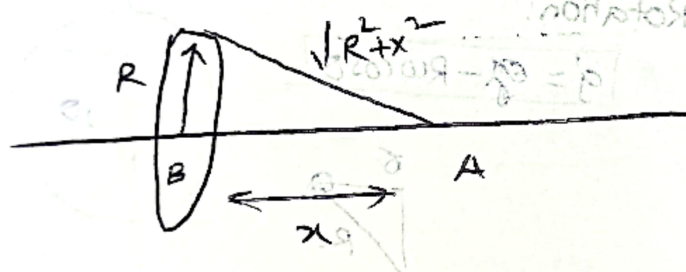
$$F_{\text{net}} = F_1 + F_2 + F_3$$

$$F_{\text{net}} = \frac{GM_1}{r_1^2} + \frac{GM_2}{r_2^2} + \frac{GM_3}{r_3^2}$$

Q. Based on Vectors:

Continuous Masses

Ring Mass: (MR)

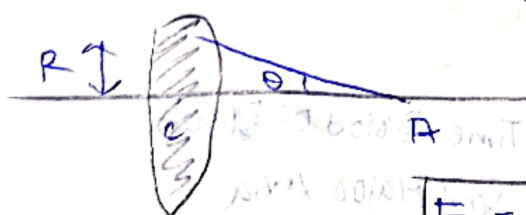


$$E_A = \frac{GMx}{(R^2 + x^2)^{3/2}}$$

$$E_C = 0$$

i. E is Max at $\frac{R}{\sqrt{2}}$

Uniform Disc (MR):



$$E_A = \frac{2GM}{R^2} (1 - \cos\theta)$$

$$E_C = 0$$

Semi Circular Ring (M, R): $\lambda = \frac{M}{\pi R}$ $E = \frac{2G\lambda}{R}$ Mass per Length

Quarter Ring:

$$E = \frac{\sqrt{2}G\lambda}{R}$$

Finite Linear Mass: λ

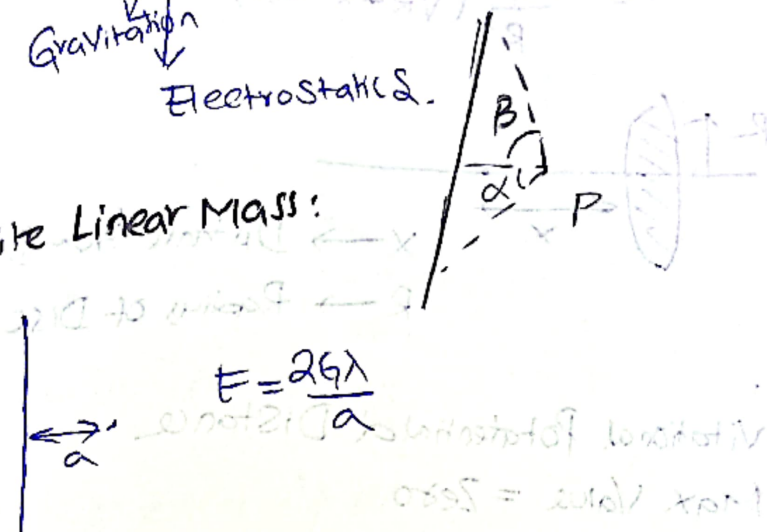
$$E_P = \frac{G\lambda}{a} [(\sin\alpha + \sin\beta)(-i) + (\cos\alpha - \cos\beta)j]$$

$$E_P = \frac{K\lambda}{a} [(\sin\alpha + \sin\beta)(-i) + (\cos\beta - \cos\alpha)j]$$

Gravitation $\downarrow$

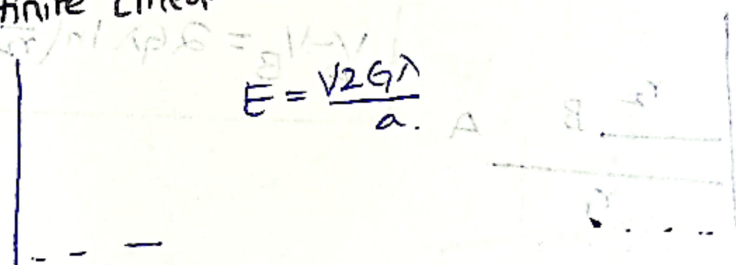
Electrostatics

Infinite Linear Mass:



$$E = \frac{2G\lambda}{a}$$

Semi Infinite Linear Mass:

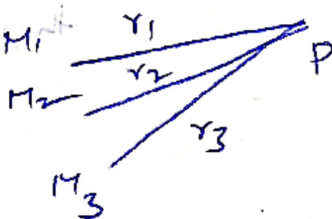


$$E = \frac{\sqrt{2}G\lambda}{a}$$

Gravitational Potential (V) Scalar Quantity.

1. $\bar{V}$ due to Point Mass: $V_P = -\frac{GM}{r}$

2. $\bar{V}$ due to no. of Point Masses:



$$V_P = -\frac{GM_1}{r_1} - \frac{GM_2}{r_2} - \frac{GM_3}{r_3}$$

Ring: $V_A = \frac{-GM}{\sqrt{R^2+x^2}}$ $V_C = \frac{-GM}{R}$

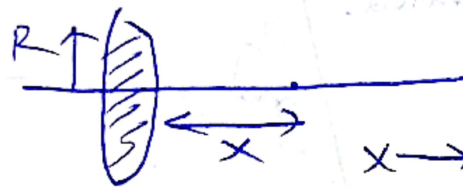
SemiCircular Ring: $V_C = \frac{-GM}{R}$

Circular Disc:

$a(MR)$

$V_C = \frac{-2GM}{R}$

$V = \frac{-2GM}{R^2} (\sqrt{R^2+x^2} - x)$

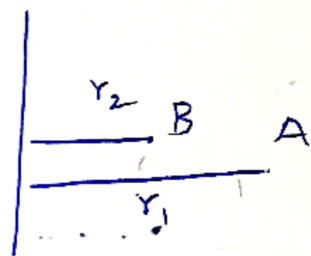


$x \rightarrow$ Distance from Disc

$R \rightarrow$ Radius of Disc.

Gravitational Potential $\propto$ Distance

Max Value = Zero.



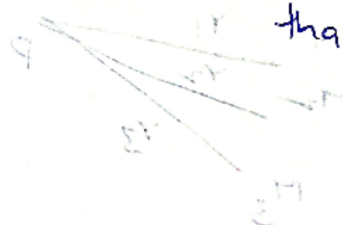
$V_A - V_B = 2G\lambda \ln\left(\frac{r_1}{r_2}\right)$

$F_M = (\bar{F} \text{ at that Point}) (m)$

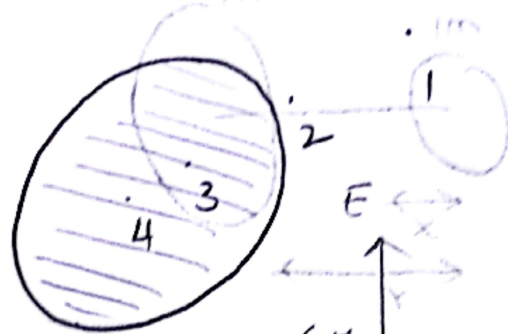
$U_M = (\bar{V} \text{ at that Point}) (m)$

Net force Acting at that Particular Point.

Net Potential Acting at that Point..



Solid Sphere.



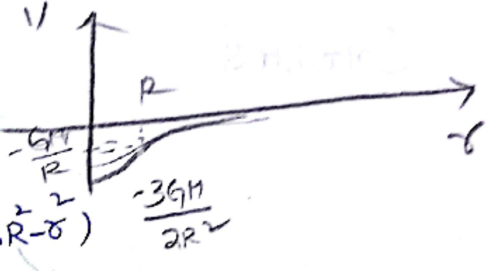
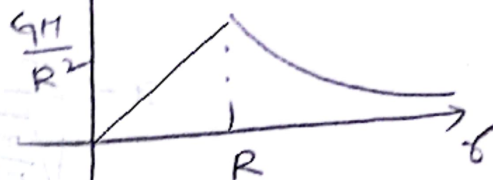
$\vec{E}$ $\vec{V}$

$$1. \quad \frac{GM}{r^2} \quad -\frac{GM}{r}$$

$$2. \quad \frac{GM}{R^2} \quad -\frac{GM}{R}$$

$$3. \quad \frac{GM}{R^3} (r) \quad -\frac{GM}{2R^3} (3R^2 - r^2) \quad -\frac{3GM}{2R^2}$$

$$4. \quad 0 \quad -\frac{3GM}{2R} \quad (9M)$$



Thin Spherical Shell:



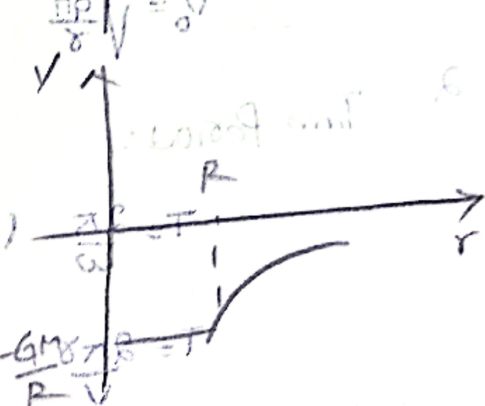
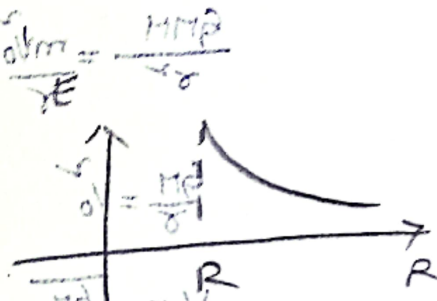
$\vec{E}$ $\vec{V}$

$$1. \quad \frac{GM}{r^2} \quad -\frac{GM}{r}$$

$$2. \quad \frac{GM}{R^2} \quad -\frac{GM}{R}$$

$$3. \quad 0 \quad -\frac{GM}{R}$$

$$4. \quad 0 \quad -\frac{GM}{R}$$

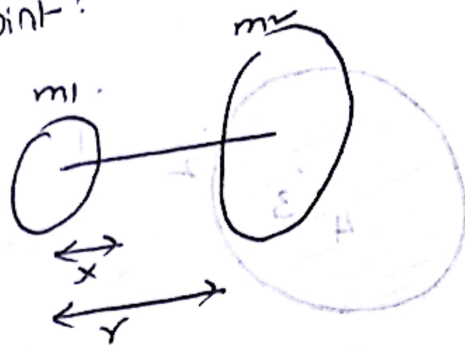


Conservation of Mech Energy:

$$\frac{GM}{r} = \frac{1}{2} v^2$$

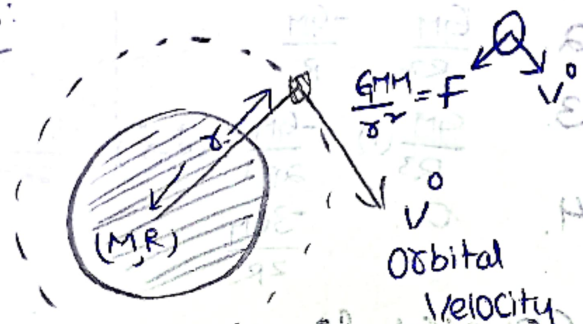
$$\boxed{K_A + U_A = K_B + U_B}$$

Null Point:



$$x = \frac{r}{1 + \frac{m_2}{m_1}}$$

Satellites:



1. Orbital Velocity:

$$\frac{GMm}{r^2} = \frac{mv_0^2}{r}$$

$$\frac{GM}{r} = v_0^2$$

$$v_0 = \sqrt{\frac{GM}{r}}$$

2. Time Period:

$$T = \frac{2\pi}{\omega} \quad (\because v = r\omega)$$

$$\omega = \frac{v}{r}$$

$$T = \frac{2\pi r}{v}$$

$$T = \frac{2\pi r}{\sqrt{\frac{GM}{r}}}$$

$$T \propto r^{3/2}$$

3. KE of Satellite :

$$K = \frac{1}{2}mv^2$$

$$K = \frac{1}{2}m\left(\sqrt{\frac{GM}{r}}\right)^2$$

$$K = \frac{GMm}{2r}$$

4. PE of Satellite :

$$U = mv$$

$$U = m\left(-\frac{GM}{r}\right)$$

$$U = -\frac{GMm}{r}$$

5. Total Energy of Satellite :

$$E = K + U$$

$$E = \frac{GMm}{2r} - \frac{GMm}{r} \times \frac{2}{2}$$

$$E = -\frac{GMm}{2r}$$

As radius $\uparrow$

$V_0 \downarrow$ $T \uparrow$ $K \downarrow$ $U \uparrow$ $E \uparrow$

Escape Velocity :



$$E_A = E_B$$

$$K_A + U_A = K_B + U_B$$

$$\frac{1}{2}mu^2 - \frac{GMm}{R} = 0 + 0 \quad \frac{1}{2}u^2 = \frac{GM}{R}$$

$$u = \sqrt{\frac{2GM}{R}} = \sqrt{2gR} \quad (\because R \text{ is radius of Earth})$$

$$u = 11.2 \text{ Km/hr}$$